INTERNATIONAL WORKSHOP AND CONFERENCE ON MATHEMATICAL MODELS FOR POST-QUANTUM CRYPTOGRAPHY AND ITS APPLICATIONS (IWCMPQC 2027) 22- 27 FEBRUARY 2027

Jointly Organized by

Centre for Mathematical Modelling and Quantum Computing(CMMQC)
Gandhigram Rural Institute
(Deemed to be University)
Tamil Nadu, India
(International Workshop)
22-24 February 2027

And

Department Of Mathematics
Cochin University Of Science And Technology
Kerala , India
(International Confernce)
25- 27 February 2027

ABOUT THE GRI

The GRI Founded by two dedicated disciples and contemporaries of Mahatma Gandhi, Dr. G Ramachandran and Dr.T S Soundaram Ramachandran, has its foundation Mahatma Ghandhi's revolutionary philosophy of Nai Talim, which effectively bridges the cognitive and practical dimensions, and focuses on the holistic development of learners, with an emphasis on skill development through value-based education, rooted in indigenous cultures and traditions.Since its establishment in 1956, GRI has been unstintingly contributing to higher education and has successfully pioneered rural higher education. The curricular model developed at GRI organically integrating teaching, research, and extension has earned the Institute, the status of Deemed University in 1976 by UGC, the five-star status by NAAC in 2002, 'A' Grade during the re-accreditation in 2010 by NAAC and 'A' Grade in its third cycle of reaccreditation in the year 2016.

ABOUT CMMQC

CMMQC is a specialized research centre at The Gandhigram Rural Institute (Deemed to be University), Dindigul, Tamil Nadu, India, functioning in association with the Department of Mathematics.
The centre focuses on interdisciplinary research in mathematical modelling, quantum computing, quantum dynamics, complex systems, and algorithmic development for emerging quantum technologies. It also promotes the application of mathematical and computational techniques to address challenging problems in science, technology, and data-driven modelling.The centre also actively contributes to academic and skill development by organizing national and international workshops, training programmes, seminars, and invited lectures. Such programmes provide researchers and students with opportunities to gain knowledge and practical exposure to recent developments in quantum computing, post-quantum cryptography, mathematical modelling, and related computational technologies.
Overall, CMMQC provides an interdisciplinary platform that brings together mathematics, computation, quantum technologies, cryptography, and artificial intelligence, fostering research, innovation, and collaboration in emerging areas of mathematical and computational science.

ABOUT CUSAT

Cochin University of Science and Technology (CUSAT) was initially constituted as the University of Cochin through an Act of Kerala Government on 10th July 1971. The CUSAT in February 1986, redefining its objectives as "promoting Graduate and Post-Graduate studies and Advanced Research in Applied Sciences, Technology, Industry, Commerce, Management and Social Sciences." CUSAT is now a world-ranking university with the specific purpose of developing higher education, emphasizing post-graduate studies and research in applied science, technology, industry, humanities, and commerce. CUSAT has consecutively been figured in the Times Higher Education World Ranking since 2017. The Times ranks around 1500 best universities worldwide annually with around 60 universities from India out of 967. CUSAT has also found a place in the QS World University Ranking and the National Institutional Ranking Framework (NIRF) of the India Government

ABOUT DEPARTMENT OF MATHEMATICS CUSAT

The combined Department of Mathematics and Statistics came into existence in November 1976, with the major objective of developing it to a Centre of Excellence in this region in mathematical sciences and applications and to provide facilities for training, study and research in these areas. This Department was bifurcated to form an independent Department of Mathematics on 1st January, 1996. The Department conducts an M.Sc. Degree programme and Ph.D. programmes. Our students find admission to Ph.D. programmes offered in various National Institutes such as IITs, IMSc, ISI, TIFR etc. Apart from giving instruction in post graduate courses, the Department has from the very beginning taken the leadership in promoting research in mathematics and its applications, especially in this part of the country. This was realized in a two-fold way one, by guiding young researchers to do their doctoral work and two, by conducting extension programmes which will improve the instruction at the school and college levels and encourage youngsters to take up research.

ABOUT IWCMPQC

The International workshop and Conference on Mathematical Models for Post-Quantum Cryptography and its Applications (IWCMPQC 2027) brings together cryptographers, mathematicians, cybersecurity engineers, computer scientists, and industrial R&D leaders.

As the digital landscape prepares for the quantum era, the primary objective of IWCMPQC 2027 is to explore how advanced mathematical domains—including Linear Algebra, Abstract Algebra, Fuzzy Logic, Functional Analysis, Probability Theory, and Numerical Methods play a role in solving complex security challenges that the post quantum era faces. The conference emphasizes next-generation cryptographic primitives and their practical deployment across Industrial Control Systems (ICS/SCADA), IoT, Smart Grids, Cloud/Edge networks, and automated manufacturing

TOPICS TO BE COVERED

- Abstract Algrbra & Algebraic Structures in Cryptography
- Lattice - based , Multivariate &Isogeny- Based Cryptography
- Linear Algebra in Spectral Threat Analysis
- Coding Theory & Error Correction
- Functional Analysis In Signal & Anomaly Detection
- Dynamical Systems & Control Theory
- Epidemiological Models (ODE/PDE Frameworks}
- Spectral Graph Theory & Algebraic Graph theory
- Combinotorial & Fuzzy Optimization
- Topological Data Analysis
- Extreme Value Thoery

CONTACT DETAILS

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